Graham Richards

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EDUCATION

University of California, Irvine — Expected Graduation: June 2027
B.S. Aerospace Engineering — 3.82 GPA | Dean's Honor List
Poway High School — 4.13 GPA | Graduated June 2024

SKILLS AND STRENGTHS

- **Leadership:** Leading and building technical teams
- **CAD and Simulation:** SolidWorks CAD and FEA, ANSYS Fluent CFD
- **Programming:** Python, Arduino, C++, Matlab, LUA
- **Fabrication:** 3D Printing, CNC Laser Cutting, hand and power tools
- **Electrical:** DC circuits, soldering
- **Construction:** Framing, finish, plumbing, electrical

WORK AND TECHNICAL EXPERIENCE

Aerodynamics Lead - Solar Airplane Research Project – UC Irvine — Fall 2024 - Present

- Mentored team members on advanced structural considerations, Solidworks CAD design and FEA workflows, Fluent CFD workflows, and woodshop fabrication principles.
- Developed a unique structural design workflow in Solidworks, improving complex procedural geometry handling and modularity, resulting in a 10x improvement in design iteration speed.
- Analyzed control authority and static stability characteristics in ANSYS Fluent CFD to guide design iterations, improving flight efficiency by 70%. Validated theoretical results using real world flight test data and Matlab to promote confidence in future aircraft design iterations.
- Designed and iterated on 3D printed construction jigs to reduce wing fabrication time, tighten assembly tolerances, and minimize accidental damages during development.
- Experimented with multiple adhesives for bonding balsa wood ribs to carbon fiber spars, including epoxy and CA glue, to determine the easiest way to meet the structural loading requirements.

Autonomous Rover CAD Design and Programing – UC Irvine — Fall 2024 - Winter 2025

- Engineered a bio-inspired mechanical claw system, allowing the rover to lift up and hold onto an aluminum can without the need for additional sensors or electronic actuators. Developed CAD models in Solidworks and 3D printed most of the components.
- Programmed Arduino embedded systems in C++ to provide efficient line following capabilities and computer-vision based object tracking and autonomous navigation.
- Won sixth place out of 50 teams in a timed autonomous navigation competition.

Miniature Robotic WALL-E – Personal Project — Summer 2025 - Present

- Designed and prototyped compact motion transfer mechanisms to fit within an 8 inch WALL-E robot. Includes a translating shoulder mechanism offering one more degree of freedom than most similar robots, for which I designed three unique cable driven actuators.
- Developed parametric 3D printed sections in Solidworks to form the body of the robot.

Team Lead – Tiny House Intensive Construction – Poway High School — Winter 2022 - Fall 2023

- Led a group of six students in flooring and plumbing installation for a trailer based tiny home. Interpreted routing diagrams and ensured the components were properly assembled throughout the structure. Earned *OSHA 10-hour Construction Safety and Health* certification

HONORS AND AWARDS

Dean's Honor List – Academic Achievement — 2024 - 2026 (6x)
Intuit Scholarship – Merit-Based Award — 2024
AP Scholar with Distinction – Academic Achievement — 2024
Excellence in Tech: Director's Award – Set Design (Theater Tech) Award — 2022 - 2024 (3x)